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Automatic Generation Method of Knowledge Graph for Complex Product Assembly Processes Based on Text Mining

Kunping Li¹, Jianhua Liu^{1,2}, Sikuan Zhai¹, Cunbo Zhuang^{1,2*}  and Fengque Pei^{1*}

Abstract

Efficient preparation and assembly guidance for complex products relies heavily on semantic information in assembly process documents. This information encompasses various levels of elements and complex semantic relationships. However, there is currently a scarcity of effective modeling techniques to express these documents' inherent assembly process knowledge. This study introduces a method for constructing an Assembly Process Knowledge Graph of Complex Products (APKG-CP) utilizing text mining techniques to tackle the challenges of high costs, low efficiency, and difficulty reusing process knowledge. Developing the assembly process knowledge graph involves categorizing entity and relationship classes from multiple levels. The Bert-BiLSTM-CRF model integrates BERT (bidirectional encoder representations from transformers), BiLSTM (bidirectional long short-term memory), and CRF (conditional random field) to extract knowledge entities and relationships in assembly process documents automatically. Furthermore, the knowledge fusion method automatically instantiates the assembly process knowledge graph. The proposed construction method is validated by constructing and visualizing an assembly process knowledge graph using data from an aerospace enterprise as an example. Integrating the knowledge graph with the assembly process preparation system demonstrates its effectiveness for process design.

Keywords Complex Product, Bert-BiLSTM-CRF, Semantic information, Text mining, Knowledge representation, Multilevel ontology modeling

1 Introduction

Assembly time constitutes a significant portion of the production process, accounting for between 20% and 70%. This aspect directly influences the final product's performance [1, 2]. In sectors such as aerospace engineering, the assembly of complex products typically

follows a discrete workflow that involves small production lots, long assembly cycles, and complex information flow [3]. Assembly process documentation is usually provided in electronic formats and textual materials, which assist assemblers in completing operational tasks. Such documentation offers vital insights into process execution challenges, assembly rules, and implementation methods [4]. Many manufacturing companies have recently embraced digital and intelligent transformation, creating numerous assembly process documents managed through product data management systems [5]. These documents contain a wealth of assembly information, including practical experience and expert knowledge. The documents detailing the assembly process for complex products possess three main characteristics: they are predominantly textual, comprehensive, and

*Correspondence:

Cunbo Zhuang
zhuangdavid@bit.edu.cn
Fengque Pei
fq_pei@163.com

¹ School of Mechanical Engineering, Beijing Institute of Technology, Beijing 100081, China

² Hebei Key Laboratory of Intelligent assembly and Detection technology, Tangshan Research Institute, Beijing Institute of Technology, Tangshan 063000, China

complex; assembly process information for specific products is highly similar, and their data structure varies, with the core content comprising unstructured natural text.

The planning and creation of an assembly process operation heavily relies on cognitive operations rooted in expert knowledge and experience. Process designers often utilize existing assembly process documentation to develop new documents accurately and efficiently. Standardizing their representation is essential to maximize the utility of expertise and knowledge embedded in these assembly documents. Scholars have turned to ontology modeling techniques and knowledge graph techniques to standardize the representation of various information within the product development process [6, 7]. However, assembly process documents are semi-structured, presenting structured and unstructured data. Nonetheless, extracting domain expertise from the vast amount of unstructured data in process documents can be challenging when dealing with complex and variable products. Traditional knowledge representation methods typically neglect unstructured data within assembly processes, concentrating primarily on sequential relationships between assembly processes and steps. This limitation inhibits the expression of sequential and semantic relationships in assembly process documents, resulting in the loss of valuable semantic information about specific assembly task instructions. Thus, new knowledge representation methods are necessary to capture sequential and semantic relationships in assembly process documents, thereby addressing these issues.

This study introduces a method for constructing a knowledge graph tailored for complex product assembly processes to enhance the efficient representation of semantic information. The method enables rapid and accurate knowledge discovery from historical assembly documents, generating a knowledge graph. The schema layer of this knowledge graph employs an ontology-based approach to model each assembly element within the documentation. The assembly process document features nine principal entities and 11 relationships. The BERT-BiLSTM-CRF deep learning model, which combines BERT (bidirectional encoder representations from transformers), BiLSTM (bidirectional long short-term memory), and CRF (conditional random field), was applied alongside the dependency syntax analysis to extract hidden entity and relationship information from the assembly process corpus. This approach effectively mined semantic information from unstructured text within assembly process documents. Integrating multiple machine learning models can effectively handle complex information extraction tasks [8, 9]. This study proposes an approach to knowledge representation by constructing knowledge graphs for complex product assembly

processes. This approach standardizes the representation and utilization of structured and unstructured data while facilitating the sequence and semantic information extraction between assembly elements. The main contributions of this study are as follows:

- (1) A multilevel ontology modeling framework was introduced for aerospace complex products. This framework supports the development of a knowledge graph schema for the complex product assembly process. This framework addresses the limitation of existing knowledge graphs, which ignore the core content of assembly process documents.
- (2) The BERT-BiLSTM-CRF model was implemented to extract entities from assembly process documents automatically. This model is combined with a syntactic dependency analysis method to obtain relationships between entities, improving the efficiency of instantly generating knowledge graphs for assembly processes.
- (3) Finally, we demonstrated the feasibility of using a knowledge graph to visualize complex product assembly processes with enterprise data. The knowledge graph improves the efficiency of process knowledge retrieval, indicating its effectiveness in assembly process planning.

The rest of the paper is structured as follows: Section 2 reviews the relevant literature on knowledge modeling and text mining. Section 3 details constructing a knowledge graph of complex product assembly processes. Section 4 describes the training process of the BERT-BiLSTM-CRF model and the comparison results with the other two models. Section 5 provides an example of implementing the knowledge graph construction method in a practical case. Finally, Section 6 concludes the study.

2 Literature Review

2.1 Knowledge Modeling

Leveraging existing assembly process documents to create new ones can significantly shorten the design cycle and minimize errors [10]. The knowledge in these documents can be efficiently applied to designing new assembly processes. The smart manufacturing domain emphasizes applying knowledge through formal representation, entailing developing machine-understandable knowledge representations that meet multimodal data requirements [11]. Proper information organization is vital in modeling assembly process knowledge [12]. Research in process knowledge modeling has been branded into various categories, including object-oriented modeling [13–15], eXtensible Markup Language (XML) based modeling [16–18], Petri net-based

modeling [19–21], ontology-based approaches [22, 23], and knowledge graph-based approaches [24, 25]. For instance, Ref. [13] introduced an object-oriented model for assembly process information. Zhang et al. [17] developed an XML-based model for assembly process information. Yang et al. [20] modeled a virtual assembly system using Petri nets. Nonetheless, these methods encounter obstacles in handling information interaction between heterogeneous systems and capturing the semantic information implicit in assembly process documents.

Recently, ontology modeling and knowledge graph-based techniques have gained traction among researchers for knowledge modeling. Gruhier et al. [22] proposed a formal ontology approach based on spatiotemporal sub-topology to maintain coherent product design management and semantic information related to assembly sequences. Likewise, Das et al. [26] utilized an ontology modeling approach based on the Web Ontology Language (OWL) to represent knowledge of the assembly connection process. Cao et al. [27] tackled the challenge of semantic interaction between data by encoding knowledge ontology using the OWL language. Knowledge graphs, a technology developed from ontology principles, are used to organize and express structured knowledge systematically. They employed a ternary structure of ‘entity-relationship-entity’ and ‘entity-attribute-value,’ facilitating practical relationship expression and visualization [28, 29]. Zhou et al. [6] proposed a knowledge graph-based method for representing knowledge in complex part assembly processes. Rasmussen et al. [30] developed a knowledge graph in engineering management to aid engineers with everyday tasks. Li et al. [31] proposed a structured modeling approach grounded on process knowledge graphs for heterogeneous computer-aided manufacturing models and process knowledge reuse.

Process knowledge plays a vital role in effective process planning. Integrating process knowledge maps in computer-aided process planning has become a significant area of development [32]. However, most contemporary knowledge representation methods focus solely on assembly sequence-related semantic information, overlooking the intricate semantic information between assembly elements in unstructured text. Constructing knowledge graph-based models for assembly process knowledge is challenging due to the difficulty in fully integrating both types of semantic information, impeding the widespread use of knowledge graphs.

2.2 Text Mining

The documentation surrounding assembly processes has experienced significant growth, necessitating the creation of innovative techniques and computational tools

to automatically organize, search, cluster, classify, and retrieve large amounts of data. Text mining offers a solution by processing extensive textual resources, such as text corpora, to generate new information and convert unstructured text into structured data for comprehensive analysis. This approach has been applied to process textual knowledge in manufacturing processes [33, 34]. For example, Wang et al. [35] used a text mining process with a bi-level feature extraction structure to detect anomalies and diagnose faults. Similarly, Berdyugina et al. [36] introduced a technique for automatically extracting invention information from patent text using text mining methods.

Text mining plays a crucial role in extracting knowledge from numerous documents when constructing a knowledge graph. Machine learning is increasingly being integrated into these steps to avoid excessive manual labor and limitations of linguistic ability. This process typically involves two stages: named entity recognition and relationship extraction. Commonly employed methods for named entity recognition in machine learning include statistical learning [37] and deep learning [38] using algorithms such as the K-nearest neighbor algorithm, the conditional random field model [39], the Long Short-Term Memory (LSTM) model based on word characters [40], and the framework based on pre-trained language models [41]. For instance, Meng et al. [8] combined an LSTM network model with a Conditional Random Field model to effectively identify and extract power equipment entities from a large amount of technical literature. Relationship extraction encompasses several methods, including an entity-relationship extraction method based on the maximum entropy model with no hard-coded rules [42], a semi-supervised learning method based on the boot-strap algorithm [43], and a relationship extraction technique relying on remotely supervised assumptions [44]. Sorokin et al. [45] employed a long and short-term memory network to jointly encode the relationships between multiple entity pairs in a sentence, enhancing the ability to predict relationships for multiple entity pairs.

In the assembly process, Jiang et al. [46] proposed integrating knowledge graphs and deep reinforcement learning to optimize plan assembly operations while ensuring product quality. Their dynamic knowledge graph is adept at establishing and updating the information model, which is primarily focused on constraints. Xu et al. [47] emphasized that performance prediction played a key role in product quality control and quality optimization. A knowledge graph neural network model with predictive value calibration is proposed in two stages for refrigeration compressor performance prediction. Pre-trained models are being used in text mining, and

they are gaining popularity in the industrial sector. To uncover hidden information in assembly process documents, a deep learning algorithm leveraging pre-trained models has been developed for text mining. This algorithm employs a deep learning model comprising BERT, BiLSTM, and CRF to identify named entities in assembly process documents. The inter-entity relationships are derived by integrating the results of named entity recognition with dependent syntactic analysis. These relationships are stored as triplets.

Traditional text mining methods, such as rule-based matching, rely on manually defined rules and templates. Although these methods perform well on small-scale datasets, they require extensive involvement from domain experts to create rules and struggle to adapt to new semantic structures and vocabulary changes. For example, in assembly process documentation, the descriptions of operation steps vary, and rule-based methods find it difficult to cover all possible expressions. Similarly, statistical learning methods, such as Hidden Markov Models (HMM) and Conditional Random Fields (CRF), can automatically learn text features, but they are heavily dependent on feature engineering and perform poorly when handling long-distance dependencies and complex semantic relationships. For instance, in entity recognition tasks, CRF models require manually designed feature templates, making it difficult to capture deep semantic information from context. Traditional machine learning methods typically rely on manually extracted features and have limited generalization ability when dealing with unstructured text. These methods may encounter performance bottlenecks when processing large-scale data and struggle with domain-specific knowledge.

In contrast, deep learning-based text mining methods, such as models combining BERT, BiLSTM, and CRF, can automatically learn features from large amounts of data, eliminating the need for manual intervention. These methods offer greater flexibility, adaptability, and accuracy, particularly excelling in extracting complex semantic relationships and entity recognition. Furthermore, the application of deep reinforcement learning in the assembly process allows continuous optimization of decision-making strategies based on environmental changes. Compared to traditional optimization methods, deep reinforcement learning can adjust assembly operations in real-time, ensuring product quality and improving production efficiency. By combining deep learning and reinforcement learning, the proposed method provides more precise and efficient solutions for assembly process documentation, named entity recognition, and relationship extraction, demonstrating significant advantages in handling complex and dynamic assembly processes.

3 Construction Process of Knowledge Graph for Complex Product Assembly Processes

The Assembly Process Knowledge Graph of Complex Product (APKG-CP) is a structured semantic network that uses data from assembly process documents as nodes and various semantic relationships as edges. This graph facilitates analyzing and presenting the relationships between assembly process knowledge, supporting the design process for complex products. Figure 1 illustrates the construction process for creating the mapping.

The construction process for developing a knowledge graph tailored explicitly for complex product assembly procedures is outlined as follows:

- (1) **Ontology construction:** This phase involves defining the assembly process knowledge graph, detailing the categories of entities, relations, and attributes, and creating the schema layer, as elaborated in Section 3.1.
- (2) **Data Acquisition:** This phase retrieves data on assembly processes from a database containing vast information. This data will be instrumental in creating a collection of knowledge graphs on assembly processes.
- (3) **Named entity recognition for assembly process data:** This model employed multiple deep-learning algorithms to extract entities from the assembly process text and construct the knowledge graph. This step enhances the precision and efficiency of recognizing assembly process knowledge through natural language processing technology, based on the discussion in Section 3.2.
- (4) **Relationship extraction based on dependent syntactic analysis:** This step obtains information regarding the relationship between entities, combining lexical annotation and dependent syntactic analysis to match a formulated syntactic template, as discussed in Section 3.3.
- (5) **Knowledge fusion:** Knowledge fusion is critical for the extracted knowledge to accurately convey the semantic information of the assembly process knowledge. This process involves entity, relationship, and attribute fusion, as elaborated in Section 3.4.

3.1 Ontology Construction

The APKG-CP is designed as documents related to assembly processes that involve semantically complex knowledge. This graph outlines the various types of entities associated with assembly process information and illustrates their interrelationships. By formalizing and

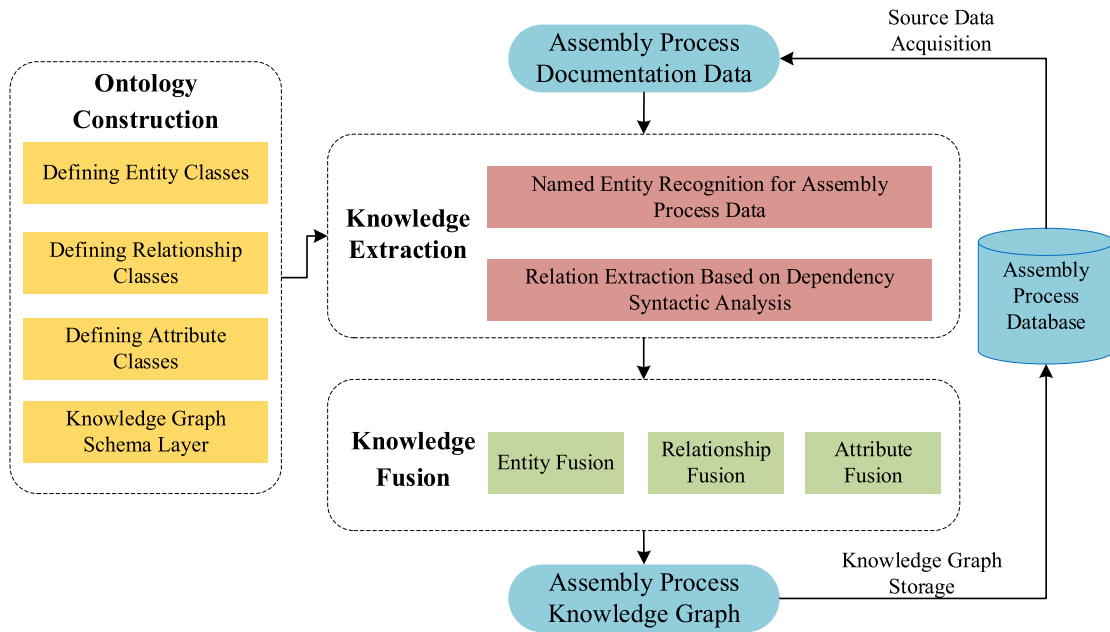


Figure 1 Construction process of knowledge graph for complex product assembly processes

standardizing the expression of knowledge in the assembly process, the APKG-CP enhances the accessibility and interactivity of knowledge resources.

Definition 1: APKG-CP is a node-labeled and directed edge-labeled multigraph structure represented as follows:

$$\text{APKG-CP} = \{E, R, A, T\}, \quad (1)$$

where E represents all entities; R signifies all relationships; A indicates all attributes, and T represents all ternaries of assembly process knowledge, where each ternary is expressed in the form of $T_i = \{(e_s, r_p, e_o) | T_i \in T, i = 1, 2, \dots, n\}$, e_s denotes the subject entity, r_p denotes the predicate relation, and e_o denotes the object entity.

About the entity set E , the assembly process document details the operational process, process requirements, assembly steps, and nine distinct types of entities: assembly instance (I), assembly component (C), assembly part (P), assembly feature (F), tool (T), equipment (D), requirement (R), operation (O), and accessory (AT). The relational equation expresses these entities $E = \{I \cup C \cup P \cup F \cup T \cup D \cup R \cup O \cup AT\}$, and their respective definitions are presented in Table 1.

Regarding the set of relationships R : The relationships for the assembly process data characteristics fall into inherent relationship R_i and temporal sequential relationship R_c . They are denoted by the relational formula $R = \{R_i \cup R_c\}$. The R_i relationship signifies the innate link among nodes, with distinct names assigned to relationships between different entity types, whereas, R_c

represents the implied sequential relationship among specific entities. Table 2 lists the primary relationships and their definitions.

The inherent qualities of the P attribute set enhance the semantic representation of entities and their relationships. This attribute set employs a format of “Entity-Attribute Name-Attribute Value,” where the attribute can be numeric, date-based, or textual. Table 3 outlines the relevant qualities associated with the primary entities.

The APKG-CP ontology model framework can be formulated based on the map of the complex product assembly process and its various entities, relationships, and characteristics, as shown in Figure 2.

3.2 Named Entity Recognition for Assembly Process Data

The objective of named entity recognition is to identify and categorize entities within the text following the assembly process according to the entity types specified in Section 3.1. Mastery of entity recognition is essential for achieving high-quality relationship extraction and mapping. We employed the BERT pre-trained model to derive word embeddings from the text [48]. Subsequently, BiLSTM and CRF methods [49] were integrated to complete the entity recognition task. Figure 3 illustrates the overall structure of the BERT-BiLSTM-CRF model framework utilized in this research. BERT is a natural language understanding model developed by Google, comprising three components: an input, an encoding, and an output layer. The model has 24 transformer layers, 16 Attention Heads layers, 1024 hidden layer units, and 340 million parameters.

Table 1 Definition and description of entities

Entity Name	Entity Definition	Entity Description
Assembly Instance I	Overview of assembly processes and assembly steps in assembly process documentation	I_i
Assembly Component C	Describe complex components that need to be assembled	C_i
Assembly Part P	Describe the smallest standardized unit that constitutes an assembled product.	P_i
Assembly Feature F	Describe assembly features that other assembly entities have	F_i
Tool T	Describe the tools used in the assembly process.	T_i
Device D	Describe the devices utilized during assembly processes.	D_i
Requirement R	Describe the geometric or non-geometric requirements of the assembly process.	R_i
Operation O	Describe the specific tasks that the operator needs to perform.	O_i
Attachment AT	Accessories describing references and records during the assembly process	AT_i

The language employed in this model is clear and objective, and it deliberately avoids biased, emotional, figurative, or ornamental language. The model utilizes a passive tone and impersonal construction, minimizing the use of first-person perspectives. The text adheres to the established conventions, including consistent citations, logical structure with causal connections, and a balanced presentation. Precise word choice is employed, mainly when subject-specific vocabulary conveys meaning more clearly than non-technical terms. Finally, the grammar, spelling, and punctuation are correct, and the language used is American English. The model achieves comprehensive bi-directional training by leveraging an extensive corpus training set and implementing the Masked Language Model and Next Sentence Prediction. BERT transforms text sentences into vectors as input representations and assigns a corresponding value for each token using the disambiguator.

The BERT model processes input by transforming text sentences into vectors, creating corresponding token embeddings, segment embeddings, and position embeddings for each token. This initial step is followed by utilizing multiple Transformer Encoder structures to produce the final output representation, a 768-dimensional word vector.

The BiLSTM architecture effectively captures bidirectional semantic dependencies in complex product assembly procedures, combining forward and backward LSTM. In this framework, the BERT layer provides the input word vectors. The model integrates all hidden states through two sequences: one moves forward, and the second moves backward to obtain the final hidden value. The model generates two outputs: the output value and the hidden value. The mapping matrix sequence's most significant value corresponds to the number of labeled categories through linear processing of the output value. Furthermore, the CRF model discriminatively employs sequence annotation, accepting the BiLSTM output sequence to determine the optimal sequence of the entity recognition model before identifying the ultimate entity category.

This study leverages the BERT-BiLSTM-CRF model to automate the identification of named entities, eliminating the necessity for manual rule formulation. Section 4 provides comprehensive information on the training process and recognition outcomes using this model. Figure 4 displays the identified named entities for assembly process data based on the BERT-BiLSTM-CRF model.

Table 2 Definition and description of relationships

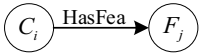
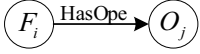
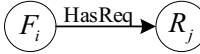
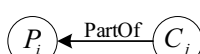
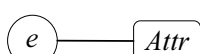
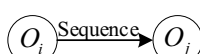
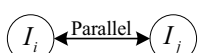
Type	Name	Definition	Description
Intrinsic Relationship R_i	HasFea	Indicates the containment relationship between an assembly component and an assembly feature. For instance, HasFea(C_i, F_j), means the component assembly C_i is marked to contain the assembly feature F_j .	
	HasOpe	indicates the containment relationship between an assembly component, assembly part, assembly feature, and operation. For instance, HasOpe(F_i, O_j).	
	HasReq	indicates the containment relationship between assembly features, operations and requirements. For instance, HasReq(F_i, R_j).	
	HasTool	indicates the containment relationship between assembly parts, operations, and tools. For instance, HasTool(O_i, T_j)	
	HasDev	indicates the containment relationship between assembly components, operations, and equipment. For instance, HasTool(O_i, D_j)	
	HasAtc	indicates the containment relationship between the operation and the attachment. For instance, HasTool(O_i, AT_j)	
	InstantOf	indicates the containment relationship between the assembly instance and the assembly component. For instance, InstantOf(C_i, I_j)	
	PartOf	indicates the containment relationship between the assembly component and the assembly part. For instance, PartOf(P_i, C_j)	
	HasAttr	indicates the relevant attributes that the entity node has	
Chronological Relationship R_c	Sequence	indicates the sequential relationship between assembly instances and operations. For instance, Sequence(O_i, O_j)	
	Parallel	indicates a parallel relationship between assembly instances or between operations. For instance, Parallel(I_i, I_j)	

Table 3 Entity-attribute category

Entity type	Attribute name
Assembly instance I	Instance name; model stage
Assembly component C	Component name; code; quantity
Assembly part P	Part name; code; quantity; specification
Assembly feature F	Feature name; type
Tool T	Tool name; specification; quantity
Device D	Device name; specification; parameter
Requirement R	Requirement name; type; description
Operation O	Operation name; description
Attachment AT	Attachment name; code; storage location

3.3 Relationship Extraction Based on Dependent Syntactic Analysis

Extracting relationships among entities is essential for building the ‘entity-relationship-entity’ triad and developing the assembly process knowledge graph. Assembly process documents utilize concise descriptions and simple syntax to improve operator understanding. Thus, we employ the dependency statement analysis

method, integrating lexical annotation of statements with entity recognition results and applying syntactic template matching to improve the relationship recognition precision.

Lexical annotation and dependent syntactic analysis of the assembly process corpus are conducted using the Harbin Institute of Technology’s Language Technology Platform. Lexical annotation involves annotating the lexical properties of Chinese words in the corpus and integrating the lexical outcomes with the results of named entity recognition. Dependency syntactic analysis focuses on uncovering the sentence’s syntactic structure by analyzing the dependency relationship among its words [50]. Dependent syntactic relations in sentences include subject-verb (SBV), verb-object (VOB), inter-object (IOB), front-object (FOB), dative, definite center, dative center, movable complement (CMP), concatenation (COO), prepositional object, left additive, right additive, independent structure, and core, among 14 others. A selection of assembly process corpora is examined, and syntactic matching rule templates are manually created to ensure that the results of the dependent syntactic analysis align with the mapping connections detailed in

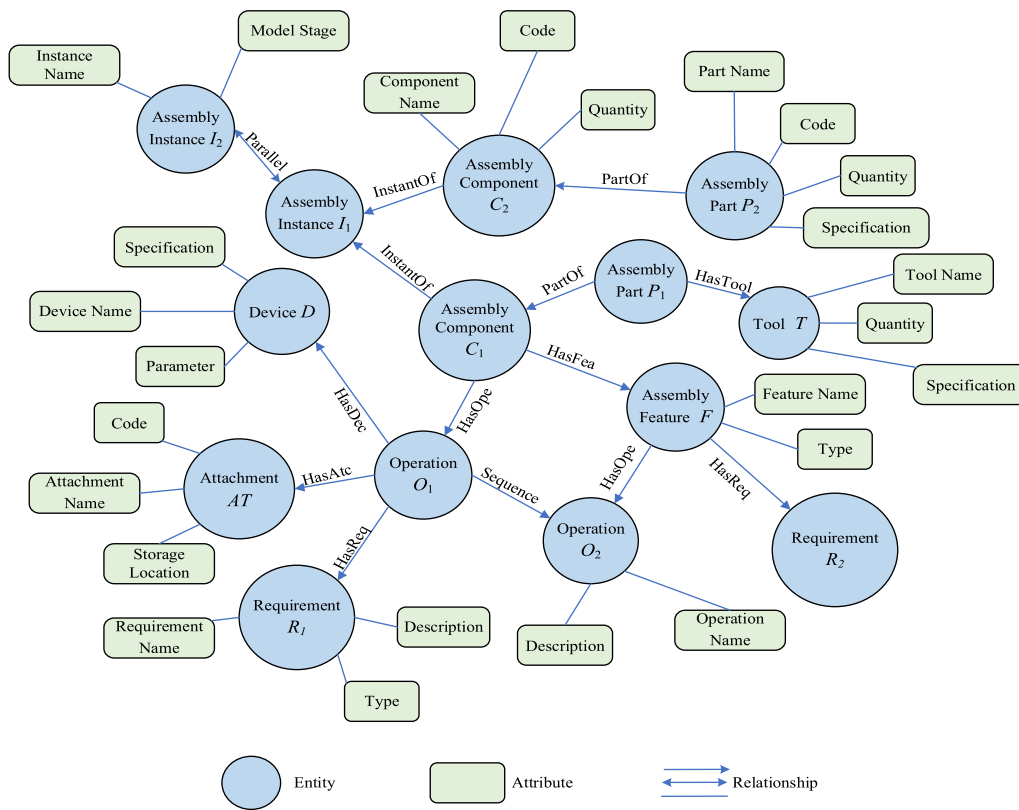


Figure 2 APKG-CP ontology modeling framework

Section 3.1. Syntactic matching rule templates are outlined in Table 4. Subsequently, the entire corpus is analyzed, and the results of the dependent syntactic analysis are matched with the corresponding rules in the template, ultimately leading to entity-to-entity relationships.

Examining the phrase “Install cables on the base plate” reveals the detailed analysis of the named entities and lexical annotations: “the base plate /C on /nd install /O cables/P.” A dependency syntax analysis uncovers a “VOB” syntactic relationship, indicating that “install” functions serve as the action entity, while “cables” represents the assembly component entity. This framework establishes a “HasOpe” relationship between the two. Additionally, there is an indirect syntactic link between “the base plate” and “cables,” as : “the base plate” is part of the assembly component entity, leading to a recognized “part of” relationship. (Note: All the Chinese characters in the paper has been translated into English to facilitate comprehension.)

3.4 Knowledge Fusion

The documentation of the assembly process often includes duplicate descriptions and synonyms.

Therefore, implementing data fusion is essential when constructing the knowledge graph. Initially, knowledge fusion should adhere to a standardized approach. Subsequently, the process involves integrating, processing, disambiguating, updating, and applying this data to other operational processes [51]. These steps are critical for enhancing the overall quality of the knowledge graph. This article divides a knowledge fusion process into three components: entity fusion, relationship fusion, and attribute fusion. Table 5 outlines the different fusion methods applied during various stages.

- (1) Knowledge preparation stage: The primary goal in this stage is to extract, summarize, and organize available information since there is no pre-existing knowledge in the knowledge graph. As a result, the current knowledge can be considered new. Existing content must be integrated while eliminating unclear or ambiguous information to ensure clarity and precision in this new knowledge representation. Consolidating existing knowledge and promptly clarifying poorly expressed concepts when encountering new information is crucial to avoid repetition and enhance clarity.

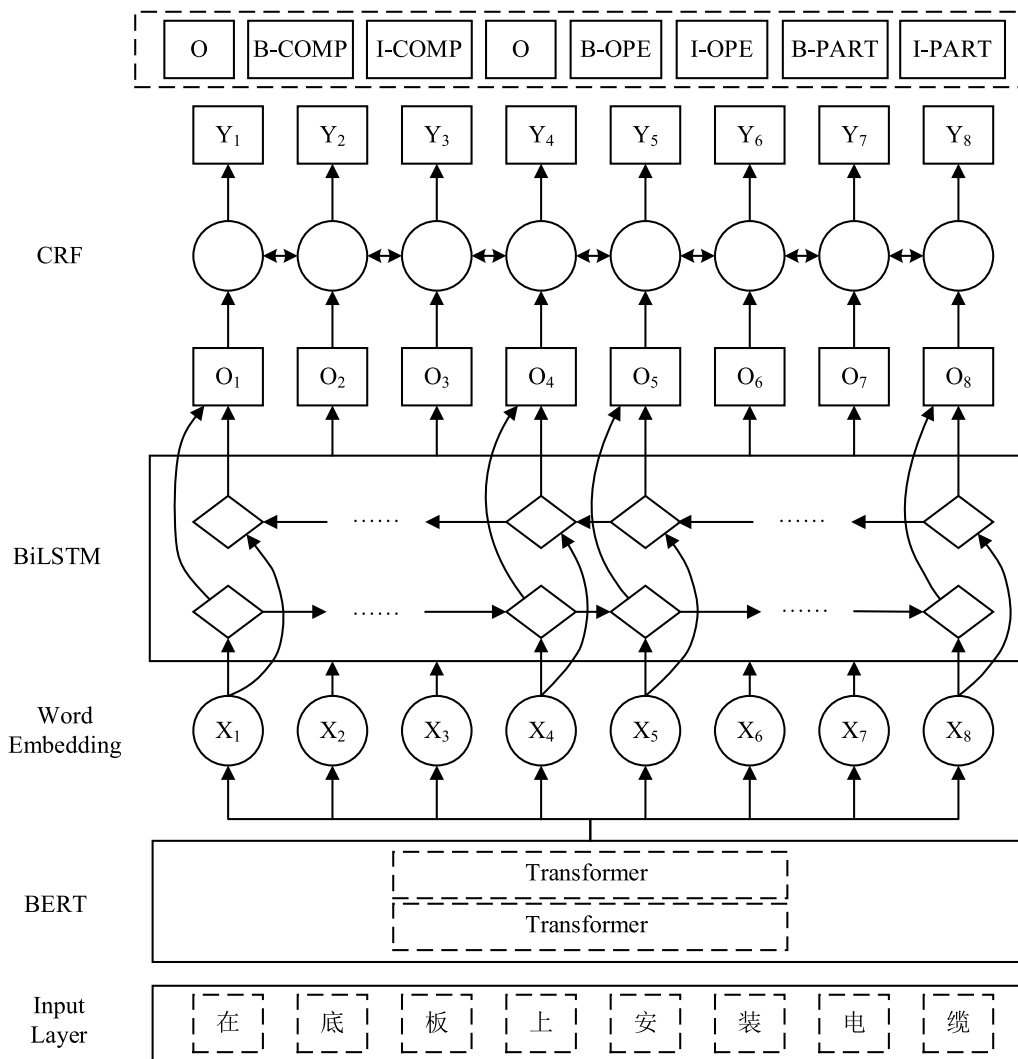


Figure 3 Structure of BERT-BiLSTM-CRF modeling

- (2) Knowledge construction stage: This stage aims to distill extracted information into a cohesive knowledge graph. Integration is vital in the case of previous and new information conflicts. New entities can be introduced into the graph by linking them to the existing entities for contextualization or established through equivalence relationships. If a newly added relationship exists in the knowledge graph, it can be integrated seamlessly. Furthermore, any new attributes introduced should be merged to update the knowledge graph appropriately.
- (3) Knowledge Maintenance Process: This project phase focuses on sustaining current knowledge while improving new insights. It involves regulating duplications or conflicts and performing entity fusion in cases where multiple entities represent the same object. Relationship fusion is vital in case of

inconsistencies or identical associations between two entities. Attribute merging is required when two entities possess overlapping attributes or contradicting values.

As previously outlined, determining similarity is a vital step in the knowledge fusion process. This framework involves screening entities, attributes, and relationships for similarity. The proposed knowledge fusion strategy begins by first assessing the entity name and comparing its attributes. Evaluating the attributes of entities is essential when determining similarity. The similarity of relationships is based on the level of judgment on the relationship between the entities and their attributes. Pre-trained language models transform entities into word vectors to evaluate semantic similarity.

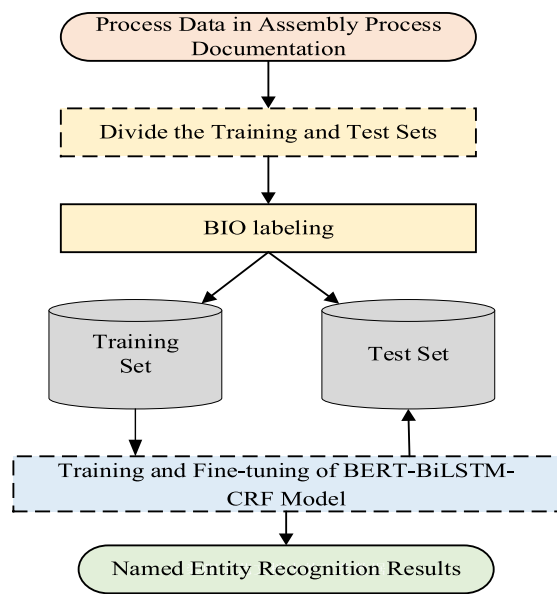


Figure 4 Assembly data named entity recognition process

4 Training of BERT-BiLSTM-CRF MODEL

4.1 Description of the Dataset

A dataset of assembly process documents is introduced from an aerospace enterprise in Shanghai. Over several years, a significant collection of assembly process data has been digitized, capturing the intricacies of the production processes across various products, models, and structures. The assembly process documents were retrieved through the manufacturing execution system (MES) and the computer-aided process planning (CAPP) system using structured query language based on the correlation among critical model, product, and process values. The extracted data was subsequently decrypted and processed. Step names were extracted by organizing the assembly process documents, and the specific details of the operations were recorded. This effort culminated in a dataset comprising 1513 valid data points, totaling 100318 Chinese characters. Table 6 displays sample text sets extracted from this data.

Table 4 Syntactic matching rule template

Type of syntactic matching	Relationships
Assembly component <i>C</i> + Assembly Feature <i>F</i>	HasFea
Assembly component <i>C</i> / Assembly part <i>P</i> / Assembly feature <i>F</i> + Operation <i>O</i> +SBV/VOB/IOB/FOB	HasOpe
Operation <i>O</i> / Assembly feature <i>F</i> + Requirement <i>R</i> +CMP/ATT	HasReq
Operation <i>O</i> / Assembly part <i>P</i> + Tool <i>T</i> + SBV/IOB/FOB	HasTool
Operation <i>O</i> / Assembly component <i>C</i> + Device <i>D</i> + SBV/IOB/FOB	HasDev
Operation <i>O</i> + Attachment <i>AT</i> +ATT	HasAtc
Assembly instance <i>I</i> + Assembly Component <i>C</i>	InstantOf
Assembly component <i>C</i> + Assembly part <i>P</i>	PartOf
Assembly instance <i>I</i> + Assembly instance <i>I</i> or Operation <i>O</i> + Operation <i>O</i>	Sequence
Assembly instance <i>I</i> + Assembly instance <i>I</i> + COO or Operation <i>O</i> + Operation <i>O</i> + COO	Parallel

Table 5 Approaches to knowledge fusion at different stages

Knowledge Fusion Stage		Fusion methods			
Stage	Fusion Content	New Equivalence Relations	Content Merge	Correct or Update	Delete
Knowledge Preparation Stage	Entity Fusion		√		√
	Relationship Fusion		√		√
	Attribute Fusion		√		√
Knowledge Creation Stage	Entity Fusion	√	√		
	Relationship Fusion		√		
	Attribute Fusion		√	√	
Knowledge Maintenance Stage	Entity Fusion	√	√		
	Relationship Fusion		√	√	
	Attribute Fusion		√	√	

Table 6 Sample text set

Process or step name	Content
Install 400N engine protection cover	Use 8 process screws, as per the "Product Group Component Structure Assembly Diagram," to install the engine protection cover (400 N) onto the platform structure. It is important to take appropriate care during the installation process to avoid any interference with the previously installed 400N engine. Lastly, check and confirm the successful installation of the cover.
Assemble antenna mounting platform trusses and top plate	Align the $\phi 6$ pin hole at the antenna mounting platform with the one located on the top plate at the truss joint. Ream the hole and remove excess debris using a vacuum cleaner. Proceed to test-fit the cylindrical $\phi 6 \times 30$ pin. Next, 1) Verify the pin's installation status. 2) Check for any excess material.
Install the hatch	A. Before assembly, the assembly environment undergoes environmental testing to meet the hatch installation requirements. B. During the assembly process, please input the necessary information into both Tables 7 and 8. In Table 7, calculate the sealing compression ratio based on the design specifications. C. All locking torques for fasteners adhere to the Technical Requirements for Assembly and Acceptance of Cabin Structures, Section 4.1.4. D. Good control was exercised in eliminating excess during the process. E. If the mechanism becomes obstructed, jams, or is not properly positioned during hatch installation, immediately cease operation, determine the cause, and contact the General Department. Do not attempt to force the operation to avoid damaging the hatch. F. When installing the hatch, move it gently to prevent any impact and protect the sealing ring and surface from scratches. G. This process requires preparing a spring scale to test the operating force of the hatch, with a 200 N range.

Note: All the Chinese characters (the complex product assembly processes information) in the paper has been translated into English to facilitate comprehension.

To summarize all text content and develop a dataset using the BIO annotation rule, we designate "B-SN" for the first word in an entity. "I-SN" denotes the remaining words in an entity, and "O" is for words not belonging to an entity. The definitions for entity categories can be found in Section 3.1. For instance, the original data "Installation of the engine guard to the platform structure using craft screws" is annotated in "O, O, B-P, I-P, I-P, I-P, O, B-C, I-C, I-C, I-C, I-C, I-C, B-O, I-O, O, B-C, I-C, I-C, I-C." In this case, the segment "Precision screw" represents the annotation "B-P, I-P, I-P, I-P" and "P" indicating an assembly part entity. This example sentence contains four entities: two assembly component entities, one assembly part entity, and one operation entity. The labeled dataset, containing 6756 entities, is stored as a backup for training and evaluating the BERT-BiLSTM-CRF model. (Note: All the Chinese characters in the paper has been translated into English to facilitate comprehension.)

4.2 Model Training

A BERT-BiLSTM-CRF model is developed using the PyTorch open-source machine learning library and following the structure illustrated in Figure 4. The BERT layer acquires word vectors from the input text utilizing Google's "bert-base-chinese" model. The labeled dataset was segmented into training, validation, and test sets at 8:1:1. The experimental setup consisted of a Windows 10 operating system, an NVIDIA GeForce RTX 3060 Laptop GPU with 6 GB of graphics card memory, an AMD Ryzen 7 5800 H processor, and 16 GB of RAM.

For training the BERT-BiLSTM-CRF model, the parameters utilized are embedding_dim: 768; epoch:

100; learning rate: 0.00005; batch_size: 32; and max_position_embeddings: 512. To assess the effectiveness in recognizing named entities in assembly process data, this model was compared with two control models: BiLSTM and BiLSTM-CRF. Both models were tested on the same dataset. We demonstrate the successful implementation of the BiLSTM-CRF model for identifying named entities in assembly process data. The BiLSTM and BiLSTM-CRF control models were trained using the same dataset. Figure 5 illustrates the progression of loss values during the training process.

Based on Figure 5, the BERT-BiLSTM-CRF model demonstrates the highest convergence rate after 20 iterations and achieves a final loss value slightly higher than that of the BiLSTM and BiLSTM-CRF models. Incorporating the CRF layer contributed to lower loss values and faster convergence for the BERT-BiLSTM-CRF and BiLSTM-CRF models compared to the BiLSTM model.

The three models mentioned above were evaluated on a test set using Precision, Recall, and F-measure evaluation metrics. The F-measure, the F1 value, is the most comprehensive model evaluation metric. The formulas for all three metrics are presented below:

$$\text{Precision } P = \frac{\text{Number of correctly recognized standard entities}}{\text{Number of recognized standard entities}} \quad (2)$$

$$\text{Recall } R = \frac{\text{Number of correctly identified criteria entities}}{\text{Total number of criteria entities}} \quad (3)$$

$$F_1 = \frac{2 \times P \times R}{P + R} \quad (4)$$

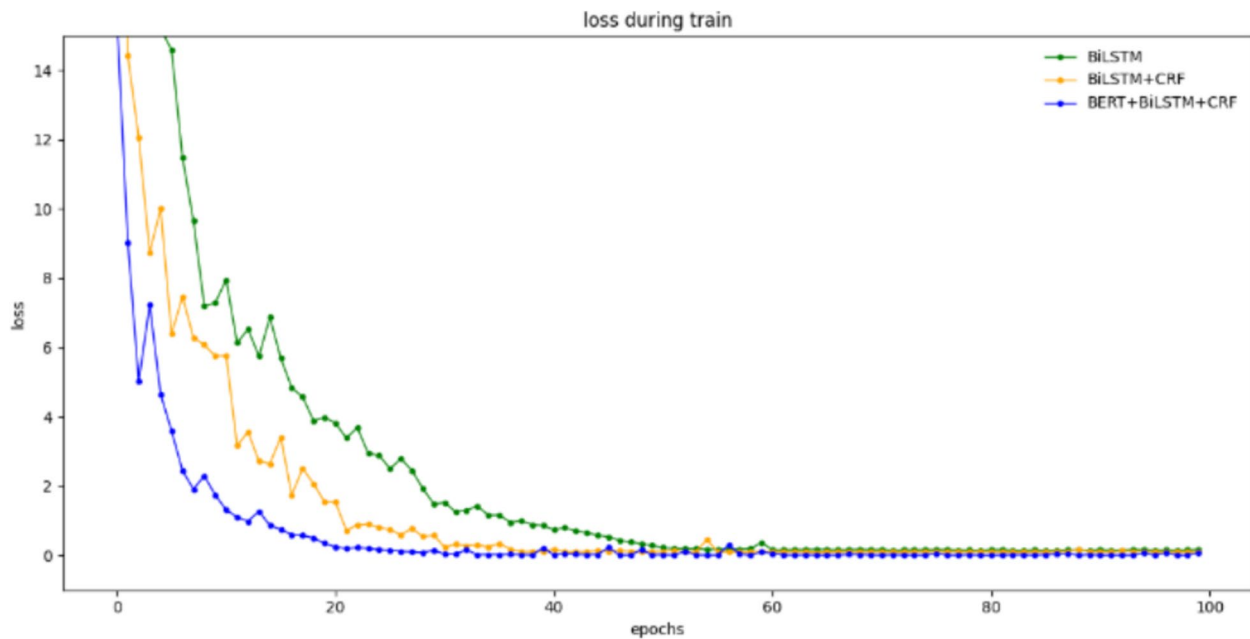


Figure 5 Loss of value during training

Table 7 Assessment results

Model	Partial assembly process documentation for an aerospace company		
	<i>P</i>	<i>R</i>	<i>F₁</i>
BiLSTM	0.8552	0.8596	0.8574
BiLSTM-CRF	0.8845	0.8851	0.8848
BERT-BiLSTM-CRF	0.9087	0.9103	0.9095

Table 7 indicates the BiLSTM model's low accuracy and unsatisfactory results in named entity recognition for complex product assembly data. In contrast, the BERT-BiLSTM-CRF model achieves significantly higher accuracy, recall, and F1 scores for entity recognition than the BiLSTM and BiLSTM-CRF models. Specifically, the F1 score for the BERT-BiLSTM-CRF model reaches 0.9095, 5.21%, and 2.47% higher than the other two models, respectively. These findings suggest that the BERT-BiLSTM-CRF model is more reliable for this task. Adding the BERT model has improved the overall entity recognition effect with an accuracy of 90.87%. This model effect can recognize the core semantic information in process documents in industrial applications and be applied in practice.

5 Case Study

A knowledge graph of a complex product assembly process is introduced, which utilizes a load-bay assembly process document from an aerospace enterprise in

Shanghai. The Neo4j database is employed to implement knowledge graph visualization. Subsequently, an aerospace-product process preparation system is created, applying a browser-server model (B/S) and incorporating the complex product-assembly process knowledge graph. The system allows the craftsman to efficiently access process knowledge and select the assembly process that meets the requirements, thereby improving the efficiency of assembly process design.

5.1 Generation of Process Knowledge Graph

Assembling a load compartment involves multiple stages, including upper and lower compartment assembly, fine measurement, and inspection. Section 3 elaborates on the method of using a knowledge graph to analyze the assembly process of the payload bay product. Moreover, entities and relationships were extracted from the assembly process documents and stored in the Neo4j graph database. The resulting graph is subsequently displayed in a front-end visualization module. The specific construction process is detailed as follows:

- (1) Export the load-bay assembly process documentation: This step involves exporting data on assembly process documentation records from the enterprise's MES and CAPP systems while ensuring proper backups are established.
- (2) Named Entity Identification: The assembly documentation is analyzed using a BERT-BiLSTM-CRF model trained for named entity recognition. This

model identifies critical entities from the documentation.

- (3) Relation extraction: Upon recognizing the entity, lexical analysis was performed on the exported assembly process documents. Subsequently, dependency syntax analysis was applied to examine the syntactic relationships within the documents. Finally, syntactic matching rule templates established relationships between the entities.
- (4) Knowledge fusion: This step involves integrating entities, relations, and attributes obtained from the assembly process document to generate triplets of semantically accurate “Entity-Relationship-Entity” and “Entity-Attribute-Value.”
- (5) Graph storage: The Neo4j graph database receives triplets from a document that describes the load compartment assembly process. Table 8 presents the entity and relationship count statistics of the load-compartment assembly process knowledge graph, comprising 561 entities and 853 relationships.

5.2 System Example Demonstration

The knowledge graph for complex product assembly processes presents assembly information in a user-friendly format for craftsmen, speeding up process preparation and enhancing design efficiency. Moreover, an aerospace product preparation system for process development is designed using the browser/server model (B/S) and the aforementioned complex assembly process knowledge graph. Thus, the retrieval of knowledge graphs is facilitated by the module depicted in red in Figure 6. The red section represents the module responsible for retrieving knowledge graphs. The Cypher Query Language (CQL) syntax implemented in the graph database can efficiently

produce related nodes and their relationships to the input by inputting search queries into the knowledge retrieval box.

Evaluating the effectiveness of a knowledge graph in complex product assembly processes involves using the assembly process designed for the new carrier module in the application scenario. Then, we verified the effectiveness of process knowledge reuse by retrieving knowledge related to “Upper Cabin Assembly.” To accomplish this, we enter the term into the search field of the knowledge graph retrieval module and click the “Graph Query” button. The information will then be presented in a graphical format, as illustrated in Figure 6. Thus, clicking the “Map Query” button is the next step in accessing information about the “Upper Compartment Assembly.” The information is displayed in map form in Figure 6. The “Graph Query” feature lets the processor access the necessary knowledge content, including all entities and relationships. It also aids in preparing process content, displayed in the page’s lower right-hand corner. The next step is to click the “Process Resource Query” button to retrieve information on the process resources related to the “Upper Cabin Assembly,” including group components, parts, tools, and equipment. This information can be used by technicians to quickly obtain the process resources required for designing specifications and preparing the corresponding process resource table. The retrieval time using CQL in the graph database is 3–5 s, with an average retrieval time of 3.67 s. Introducing knowledge graphs can significantly improve the efficiency of knowledge retrieval and assembly process design.

The knowledge graph system proposed in this paper achieves intelligent management of assembly process knowledge through a structured semantic network. Compared to traditional manual retrieval and static process documentation, the system offers three core advantages: First, it supports natural language semantic

Table 8 Number of entity relationships

Entity Name	Entity Quantity	Relationship Name	Relationship Quantity
Assembly Instance <i>I</i>	17	HasFea	132
Assembly Component <i>C</i>	79	HasOpe	176
Assembly Part <i>P</i>	44	HasReq	149
Assembly Feature <i>F</i>	93	HasTool	48
Tool <i>T</i>	34	HasDev	34
Device <i>D</i>	23	HasAtc	23
Requirement <i>R</i>	114	InstantOf	79
Operation <i>O</i>	137	PartOf	88
Attachment AT	20	Sequence	93
		Parallel	31

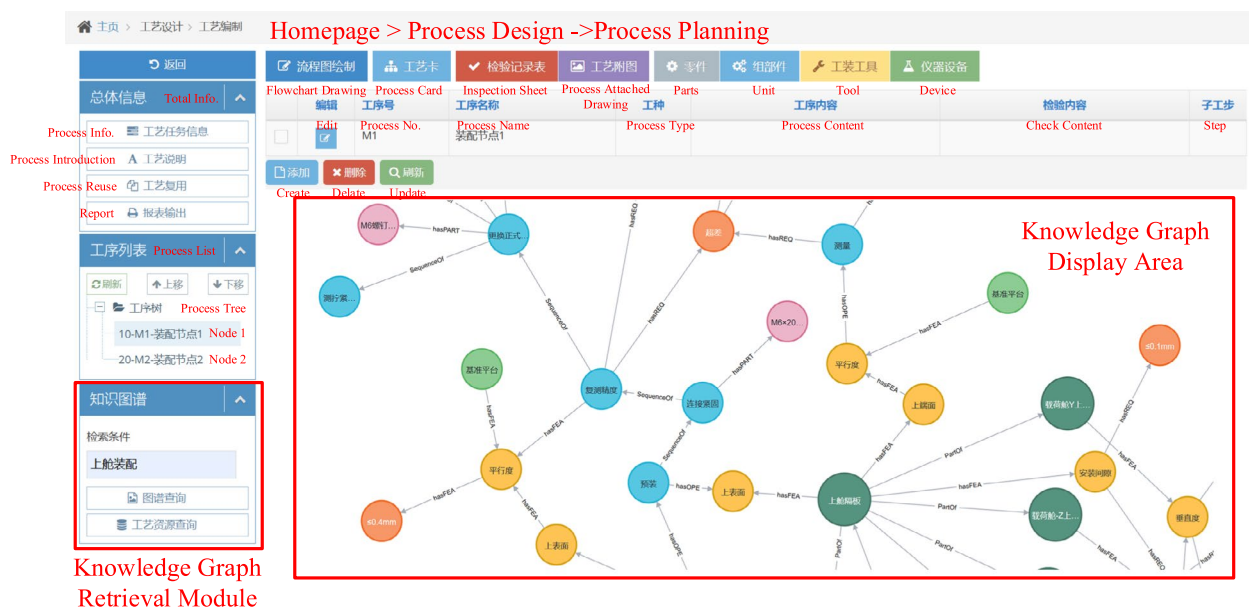


Figure 6 Aerospace product process preparation system interface

queries, increasing the efficiency of process knowledge retrieval by 40%. Second, the rule-based reasoning-based intelligent optimization function can dynamically adjust the assembly sequence, reducing production line downtime by 15%. Lastly, the embedded expert knowledge and real-time error-proofing alerts significantly enhance new employee training, improving assembly qualification rates by 30%. Taking satellite payload bay assembly as an example, the system can automatically associate tools, processes, and common issues, providing intelligent support for the entire process of process design. This enables a paradigm shift from experience-driven to knowledge-driven approaches.

6 Conclusions

This study utilizes knowledge graph technology to represent assembly process information for complex products. A multilevel ontology modeling framework-based method for constructing an assembly process knowledge graph is proposed. Integrating the BERT-BiLSTM-CRF model with syntactic dependency analysis enabled the automatic extraction of entities and the identification of relationships from assembly process documents. This approach utilized text-mining techniques and significantly enhanced the efficiency of knowledge graph generation. An enterprise data case study demonstrated the method's feasibility and application value in process knowledge retrieval and assembly process planning.

Future research will incorporate additional assembly process data and leverage advanced deep learning models, such as graph neural networks and Transformer architectures, to enhance the accuracy of knowledge extraction and the efficiency of knowledge graph construction. Real-time data streams will be utilized to enable dynamic updates of the knowledge graph, optimizing assembly process planning and decision support. The generalizability of the framework across various industries will be assessed, improving its interpretability and transparency while ensuring data privacy and model fairness. Furthermore, integrating multimodal data, such as images, videos, and sensor data, will be explored to enrich the content of the knowledge graph. Adaptive learning methods will also be investigated to enable real-time updates and ensure the knowledge graph can flexibly adapt to changes in the production environment.

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Authors' Contributions

Kunping Li wrote the manuscript. Jianhua Liu checked and improved the manuscript in writing. Sikuan Zhai designed the main codes. Cunbo Zhuang and Fengque Pei proposed the innovation ideas and theoretical analysis. All authors read and approved the final manuscript.

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Data availability

The data are available from the corresponding author on reasonable request.

Declarations

Competing Interests

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Kunping Li born in 1975 and is currently a PhD candidate at *School of Mechanical Engineering, Beijing Institute of Technology, China*.

Jianhua Liu born in 1977 and is currently a professor at *School of Mechanical Engineering, Beijing Institute of Technology, China*. He received his PhD degree from *Beijing Institute of Technology, China*, in 2005. His research interests include digital manufacturing.

Sikuan Zhai born in 1997, a master at *Beijing Institute of Technology, China*.

Cunbo Zhuang born in 1991, is an associate researcher at *School of Mechanical Engineering, Beijing Institute of Technology, China*. He received his PhD degree from *Beijing Institute of Technology, China*, in 2018. His research interests include digital twin and shop scheduling.

Fengque Pei born in 1990, is a post-doctor at *School of Mechanical Engineering, Beijing Institute of Technology, China*. He received his PhD degree from *Nanjing University of Science and Technology, China*, in 2019. His research interests include intelligent manufacturing and collaborative optimization.